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SUMMARY

Third-year B.Tech Computer Science (AI & ML) student who builds production-shaped systems, not demos. Experienced in Python and FastAPI backends, React/TypeScript frontends, and applied machine learning, with a focus on architecture, testing, and making model output verifiable. Seeking a Software Engineering, Machine Learning, or AI Systems internship.

TECHNICAL SKILLS

Languages: Python, Java, JavaScript, TypeScript, SQL, C

Backend: FastAPI, Node.js, Express, REST APIs, JWT, Celery, SQLAlchemy, Alembic

Frontend: React, Next.js, Vite, Tailwind CSS, HTML, CSS

Databases: PostgreSQL, MySQL, MongoDB, SQLite, Neo4j, Prisma

AI / ML: Scikit-learn, TensorFlow, PyTorch, Pandas, NumPy, spaCy, NLP, RAG pipelines, LLM integration, AI agents, knowledge graphs, NetworkX

Tools & Infra: Git, GitHub, Docker, Vercel, Render, pytest, Jupyter, Ollama, VS Code

Concepts: Data structures & algorithms, DBMS, operating systems, system design, modular architecture, graph algorithms, dependency analysis

PROJECTS

RedBlue / BLACKBOX — Security Layer for AI Agents | Python, FastAPI, NetworkX, React, TypeScript

- Built a system that detects indirect prompt-injection attacks (OWASP LLM01) against tool-using AI agents by reconstructing an execution graph from agent events and computing reachability-based blast radius.
- Designed a deterministic security engine as the source of truth and restricted the LLM to explanation only; model output is schema-validated and provenance-checked against real event IDs, with an offline fallback so the system never depends on LLM availability.
- Implemented a counterfactual what-if engine that re-runs the full detection stack per candidate defense to select the minimum effective intervention, then verifies it by replaying the attack against a live target agent with runtime enforcement.
- Wrote 335 automated tests with zero network calls in the suite; delivered a React dashboard for execution graph, blast radius, and verification. *Two-person hackathon team (HackWave 3.0).*

SUSTAINA — Modular Clinic Operating System | FastAPI, PostgreSQL, RAG, React, Docker

- Engineered a FastAPI backend as additive, API-compatible modules with a versioned /api/v1 router layer, so new capabilities ship without breaking existing endpoints.
- Built a RAG retrieval pipeline with graceful degradation: falls back to retrieved evidence without an LLM key, a file-backed store without a vector DB, and a hashing embedder without an embeddings model — the full clinical flow runs on Python and PostgreSQL alone.
- Modeled clinical and admin domains in SQLAlchemy with Alembic migrations (no destructive create_all), plus a tamper-evident audit log and a React + TypeScript doctor portal.

CodeLens AI — Codebase Knowledge Graph & Dependency Analysis | Python, AST, Neo4j, Knowledge Graphs, FastAPI

- Building a developer-intelligence system that models a repository explicitly — repository parsing, AST analysis, and dependency extraction into a Neo4j knowledge graph — so code questions are answered from the actual dependency graph instead of raw text.
- Structured the pipeline as independently evaluable stages (parse, AST, dependencies, graph, reasoning, visualization) with accuracy evaluation built into the architecture rather than bolted on; Milestone 1 complete, later stages in active development.

CareerLens — Explainable Resume-to-Job Matching | Python, NLP, spaCy, Full-Stack

- Shipped a system that converts a resume into ranked job matches and a prioritized learning roadmap, where every match score is explainable rather than a black-box number.
- Implemented a roadmap engine that ranks the highest-impact missing skills, turning match results into an actionable next step for the user.

AI Forest Fire & Smoke Detection | Python, Deep Learning, Computer Vision

- Shipped an end-to-end computer-vision pipeline for detecting fire and smoke in imagery, covering data preparation, model training, and evaluation.

EDUCATION

Vardhaman College of Engineering, Hyderabad | 2024 – 2028

B.Tech in Computer Science and Engineering (Artificial Intelligence & Machine Learning) | CGPA: 8.45 / 10

- Relevant coursework: Data Structures & Algorithms, DBMS, Operating Systems, Machine Learning.
- Actively preparing for GATE Computer Science 2027; practicing DSA in Java alongside coursework.